

Database table creation:

Step1 : Create a Schema

In MySQL Workbench:

1. Open MySQL Workbench and connect to your MySQL server.
2. In the left panel, find **Navigator** → **SCHEMAS**.
3. Right-click inside the Schemas area and select **Create Schema**.
4. Enter a name, for example: airline_project
5. Click **Apply**.
6. Review the SQL command: CREATE SCHEMA `airline\_project`;
7. Click **Apply**, then **Finish**.
8. Refresh the Schemas panel.
9. Select the new schema as the active database: USE airline_project;

Step 2 : Create a staging table

1. Open a new SQL tab in MySQL Workbench.
2. Select your schema: USE airline_project;

```
CREATE TABLE staging_flights (  
  
    YEAR VARCHAR(100),  
  
    QUARTER VARCHAR(100),  
  
    MONTH VARCHAR(100),  
  
    DAY_OF_MONTH VARCHAR(100),  
  
    DAY_OF_WEEK VARCHAR(100),  
  
    FL_DATE VARCHAR(100),  
  
    OP_UNIQUE_CARRIER VARCHAR(100),  
  
    OP_CARRIER_AIRLINE_ID VARCHAR(100),  
  
    TAIL_NUM VARCHAR(100),  
  
    OP_CARRIER_FL_NUM VARCHAR(100),  
  
    ORIGIN_AIRPORT_ID VARCHAR(100),  
  
    ORIGIN VARCHAR(100),
```

ORIGIN_CITY_NAME VARCHAR(100),
ORIGIN_STATE_ABR VARCHAR(100),
ORIGIN_STATE_NM VARCHAR(100),
DEST_AIRPORT_ID VARCHAR(100),
DEST VARCHAR(100),
DEST_CITY_NAME VARCHAR(100),
DEST_STATE_ABR VARCHAR(100),
DEST_STATE_NM VARCHAR(100),
CRS_DEP_TIME VARCHAR(100),
DEP_TIME VARCHAR(100),
DEP_DELAY VARCHAR(100),
DEP_DELAY_NEW VARCHAR(100),
DEP_DEL15 VARCHAR(100),
TAXI_OUT VARCHAR(100),
WHEELS_OFF VARCHAR(100),
WHEELS_ON VARCHAR(100),
TAXI_IN VARCHAR(100),
CRS_ARR_TIME VARCHAR(100),
ARR_TIME VARCHAR(100),
ARR_DELAY VARCHAR(100),
ARR_DELAY_NEW VARCHAR(100),
ARR_DEL15 VARCHAR(100),
CANCELLED VARCHAR(100),
CANCELLATION_CODE VARCHAR(100),

DIVERTED VARCHAR(100),
CRS_ELAPSED_TIME VARCHAR(100),
ACTUAL_ELAPSED_TIME VARCHAR(100),
AIR_TIME VARCHAR(100),
DISTANCE VARCHAR(100),
CARRIER_DELAY VARCHAR(100),
WEATHER_DELAY VARCHAR(100),
NAS_DELAY VARCHAR(100),
SECURITY_DELAY VARCHAR(100),
LATE_AIRCRAFT_DELAY VARCHAR(100),
DIV_AIRPORT_LANDINGS VARCHAR(100),
DIV_REACHED_DEST VARCHAR(100),
DIV_ACTUAL_ELAPSED_TIME VARCHAR(100),
DIV_ARR_DELAY VARCHAR(100),
DIV_DISTANCE VARCHAR(100),
DIV1_AIRPORT_ID VARCHAR(100),
DIV1_TOTAL_GTIME VARCHAR(100),
DIV2_AIRPORT_ID VARCHAR(100),
DIV2_TOTAL_GTIME VARCHAR(100),
DIV3_AIRPORT_ID VARCHAR(100),
DIV3_TOTAL_GTIME VARCHAR(100),
DIV4_AIRPORT_ID VARCHAR(100),
DIV4_TOTAL_GTIME VARCHAR(100),
DIV5_AIRPORT_ID VARCHAR(100),

```
DIV5_TOTAL_GTIME VARCHAR(100)

);
```

Step 3: Load the data from csv

- LOAD DATA LOCAL INFILE '/Users/YourName/Downloads/airline_ontime_2026_01.csv'
- INTO TABLE staging_flights
- FIELDS TERMINATED BY ','
- ENCLOSED BY '"'
- LINES TERMINATED BY '\n'
- IGNORE 1 LINES;

Step 4 : If you get error as loading is disabled:

- Check the settings : SHOW VARIABLES LIKE 'local_infile';
- If the value is OFF run : SET GLOBAL local_infile = 1;
- Now disconnect and reconnect to MySQL Workbench.

Step 5 : If SET GLOBAL gives an access error

In MySQL Workbench:

1. Return to the Workbench home screen.
2. Right-click your MySQL connection.
3. Select **Edit Connection**.
4. Open the **Advanced** tab.
5. In the **Others** field, add: OPT_LOCAL_INFILE=1
6. Click **Test Connection**.
7. Click **OK**.
8. Reconnect to the database.

Run the Import load query again

Step 6 : Check the count of all months . Result should be as follows:

```
SELECT `MONTH`, COUNT(*) AS row_count FROM staging_flights WHERE `MONTH` = '1'

GROUP BY `MONTH`;
```

1. 562439
2. 515037
3. 612102
4. 597919
5. 611735
6. 607577

Step 7 : Now the raw table is ready and we will normalise the raw file into 13 tables and have our raw file data as a copy.